Yuzhu Tao

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Research Interests

Computer Vision, Image Stitching, Image Editing Models, Multimodal Large Language Models, Event-based Vision.

Education

M.Sc. Beijing Jiaotong University, Artificial Intelligence — Beijing, China, Sept 2024 – June 2027
- Research Focus: Computer Vision, Image Stitching, Geometric Modeling, Event-based Vision, Image Editing, and Multimodal Large Models.
- Honors: First-Class Graduate Scholarship twice, Outstanding Student Cadre, Outstanding Graduate CPC Member.

B.Eng. Hebei Normal University, Computer Science and Technology — Hebei, China, Sept 2020 – June 2024
- GPA: 3.83/4.00; Rank: 1/147.
- CET-6: 453.
- Honors: Provincial Outstanding Graduate; Lanqiao Cup C/C++ Provincial Second Prize; National College Mathematics Competition Provincial Third Prize.

Publications

Panoramic HDR Color Image from One-Second Spike Acquisition — Jan 2026
Yuzhu Tao
(Under Review at ACM Multimedia, CCF-A)

Seam-Guided Unsupervised Image Stitching with Parallax-Aware Mask Generation — Jan 2026
Yuzhu Tao
(IEEE Transactions on Circuits and Systems for Video Technology, CCF-B)

Experience

Baidu Inc., Multimodal Algorithm Research Intern — China, Mar 2026 – present, 5 months
- Conduct algorithm research on image understanding, image editing, parameter prediction, and generative model optimization.
- Built a structured dataset for image-editing operators and editing instructions.
- Fine-tuned Qwen3-VL-4B with LoRA for intelligent photo-editing parameter prediction.
- Designed a two-stage training strategy to extend the model from single-image understanding to before-and-after editing difference modeling.
- Achieved an F1-score of 92% on the test set.
- Developed pose-aware alignment and structure-aware completion methods for AI portrait outfit replacement.

Projects

Panoramic Image Stitching for Equipment Rooms Based on Spherical Projection — Beijing, China
Core Algorithm Engineer — Sept 2024 – Sept 2025
- Proposed a fast panoramic stitching algorithm based on spherical projection for equipment-room scenarios with limited single-camera viewpoints.
- Achieved panoramic reconstruction from 30 images within one second.
- Estimated horizontal focal length using a cylindrical projection model and obtained dynamic FOV with zoom-lens information.
- Built a spherical projection model and generated equirectangular projection panoramas through linear projection blending.
- Deployed the algorithm in a real-world China Tower business scenario.

Patents

Chinese Invention Patent for HDR Panoramic Color Image Reconstruction Jan 2026

- Patent No.: 2026100594905.
- Related to panoramic HDR color image reconstruction from spike acquisition and generative colorization.

Awards

First-Class Graduate Scholarship, Beijing Jiaotong University, twice.

Outstanding Student Cadre, Beijing Jiaotong University.

Outstanding Graduate CPC Member, Beijing Jiaotong University.

Provincial Outstanding Graduate, Hebei Province.

Lanqiao Cup C/C++ Programming Competition, Provincial Second Prize.

National College Mathematics Competition, Provincial Third Prize.

Skills

Programming: Python, C++

Deep Learning: PyTorch, Transformers, Diffusers, LoRA

Computer Vision: OpenCV, Image Stitching, HDR Imaging, Event-based Vision, Image Editing

Multimodal Models: Qwen-VL, Qwen-Image-Edit, Gemini, SAM

Languages: Chinese (native), English (CET-6)